

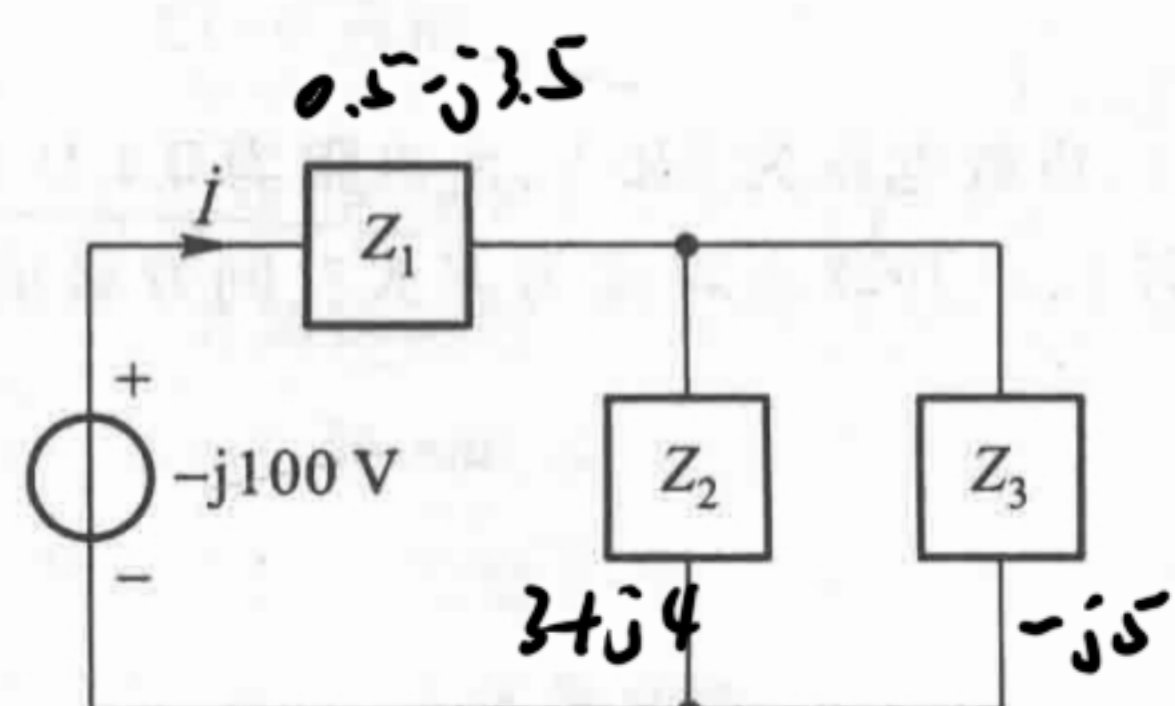
电分作业第九章

9-7 电压 $u(t) = 10\cos(100t) - 20\cos(100t + 30^\circ)$ V 施加于 $10\ \Omega$ 电阻, 求电阻的平均功率。

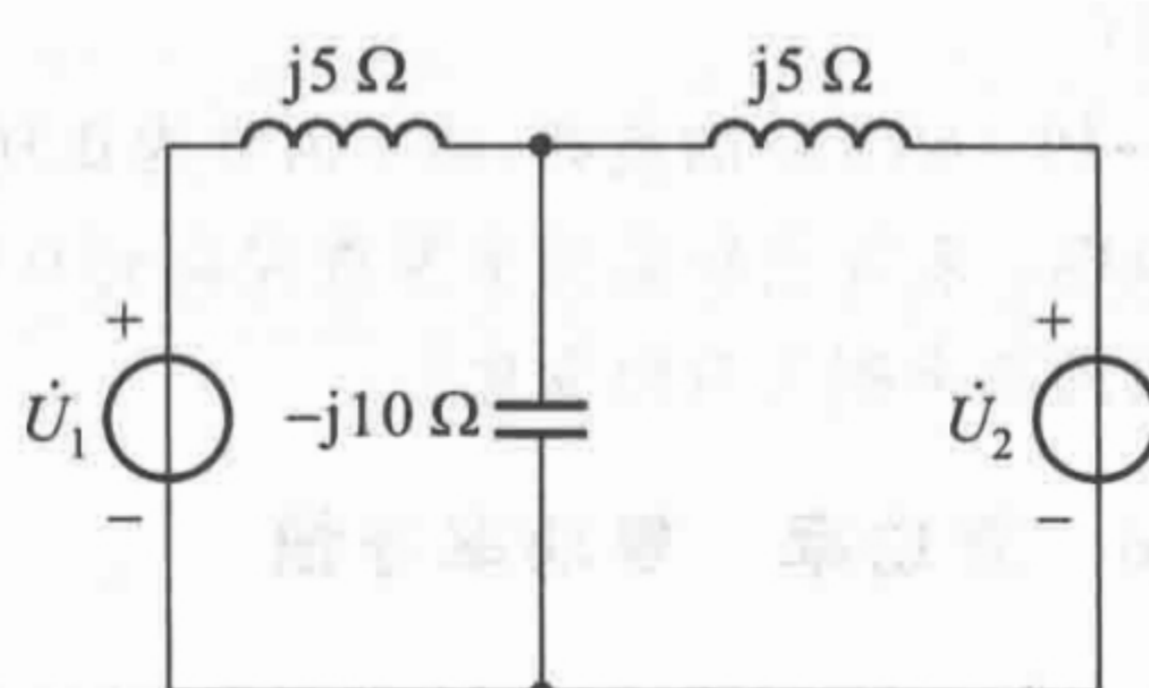
$$\begin{aligned} U_m &= 10\angle 0^\circ - 20\angle 30^\circ \\ &= 10 - 20\left(\frac{\sqrt{3}}{2} + j\frac{1}{2}\right) \\ &= 12.39\angle -126.21^\circ \end{aligned}$$

$$P = \left(\frac{U_m}{\sqrt{2}}\right)^2 \frac{1}{R} = 7.6795\text{ W}$$

9-10 电路如图题 9-8 所示。(1) 求 $\dot{I}$; (2) 求整个电路吸收的平均功率和功率因数。已知 $Z_1 = (0.5 - j3.5)\ \Omega$ 、 $Z_2 = 5\angle 53^\circ\ \Omega$ 、 $Z_3 = 5\angle -90^\circ\ \Omega$, 电源电压有效值相量为 $100\angle -90^\circ$ V。



图题 9-8



图题 9-9

$$Z = 0.5 - j3.5 + \frac{(3 + j4) \times (-j5)}{3 + j4 - j5} = 8 - j6 = 10\angle -36.9^\circ$$

$$\dot{I} = \frac{\dot{U}}{Z} = \frac{-j100}{8 - j6} = 10\angle -53.13^\circ$$

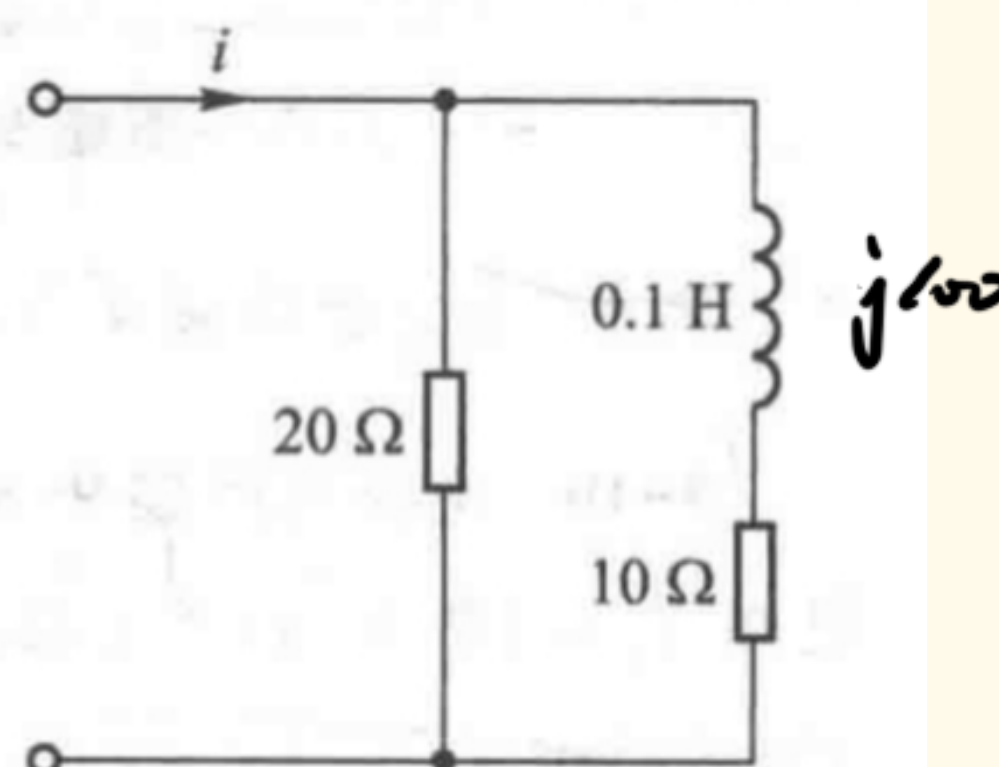
$$P = I^2 R = 10^2 \cdot 8 = 800\text{ W}$$

$$\lambda = \cos\phi = \cos(-36.9^\circ) = 0.8 \text{ (电感性)}$$

9-16 电路如图题 9-12 所示, 已知电流 i 的有效值 $I = 10\text{ A}$, $\omega = 1000\text{ rad/s}$ 。试求电路的 P 、 Q 和 S 。

9-17 某负载需功率 40 kW 功率因数为 0.84 (感性), 负载电压有效值相量为 $220\angle 0^\circ$ V。已知输电线的阻抗为 $(0.1 + j0.25)\ \Omega$ 。(1) 试求输电线的功率和无功功率; (2) 试求电源提供的功率和无功功率。

9-18 某负载的平均功率及无功功率为 6 W 及 8 var , 已知负载电压 $u(t) = 100\cos(100t)$ V, 如要求负载功率因数为 1 应并联多大的电容?



图题 9-12

9-16

$$Z = \frac{20 \cdot (10 + j100)}{20 + 10 + j100} = 18.9 + j3.67$$

$$P = I^2 R = 1890\text{ W}$$

$$Q = I^2 X = 367\text{ var}$$

$$S = \sqrt{P^2 + Q^2} = 1925.3 \text{ V}\cdot\text{A}$$

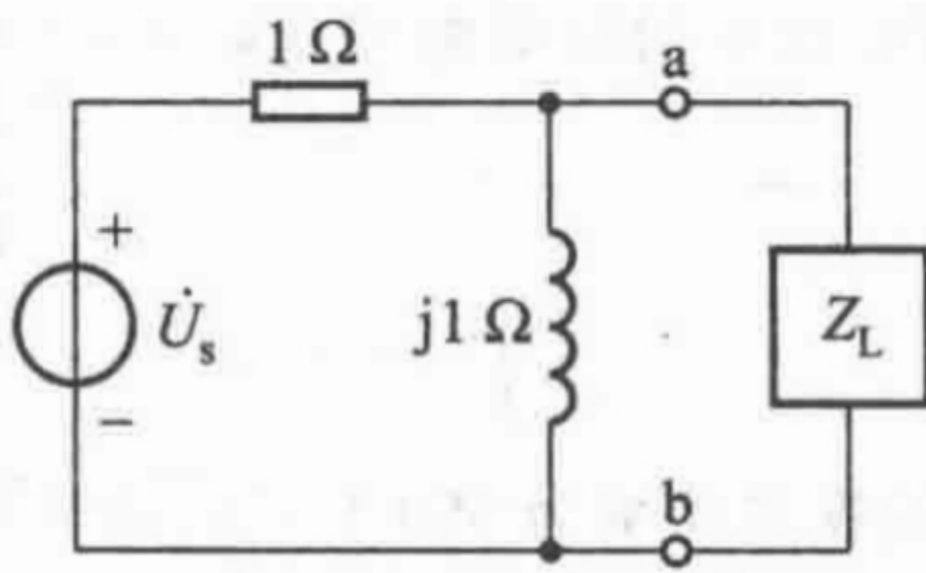
9-18.

$$Q = Q_L + Q_C = 0$$

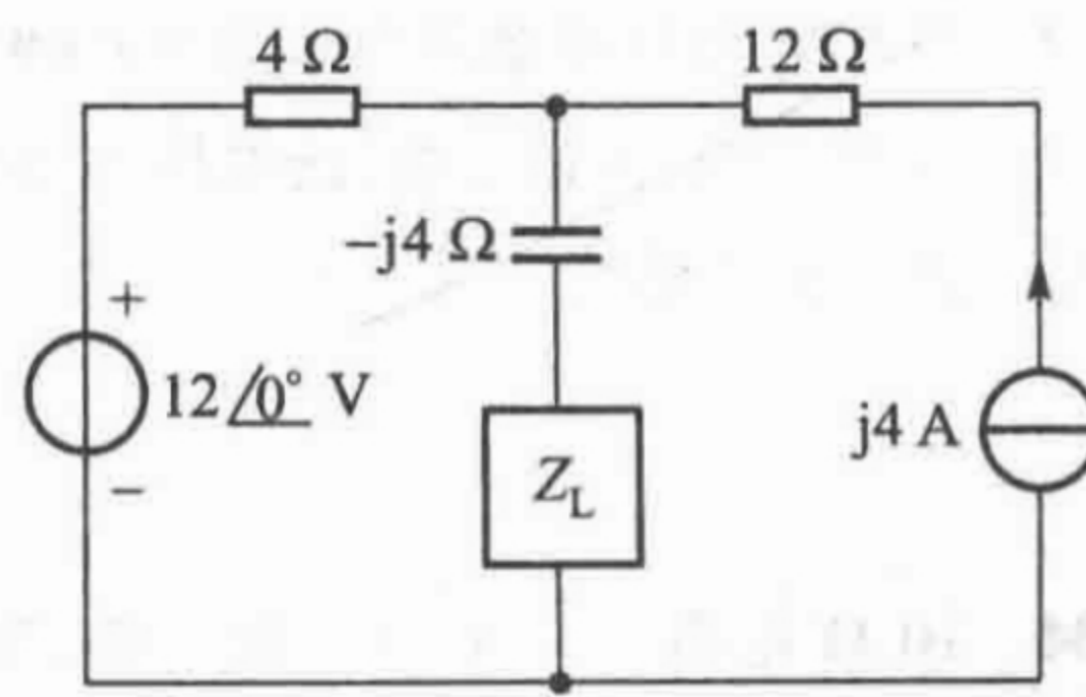
$$Q_C = -Q_L = -8 \text{ var}$$

$$Q_C = -\omega C U^2 \Rightarrow C = -\frac{Q_C}{\omega U^2} = \frac{8}{100 \times (50\sqrt{2})^2} = 16 \mu\text{F}$$

9-24 电路如图题 9-14 所示, 若 Z_L 的实部、虚部均能变动, 若使 Z_L 获得最大功率, Z_L 应为何值, 最大功率是多少? 已知 $\dot{U}_s = 14.1 \angle 0^\circ \text{ V}$ 。



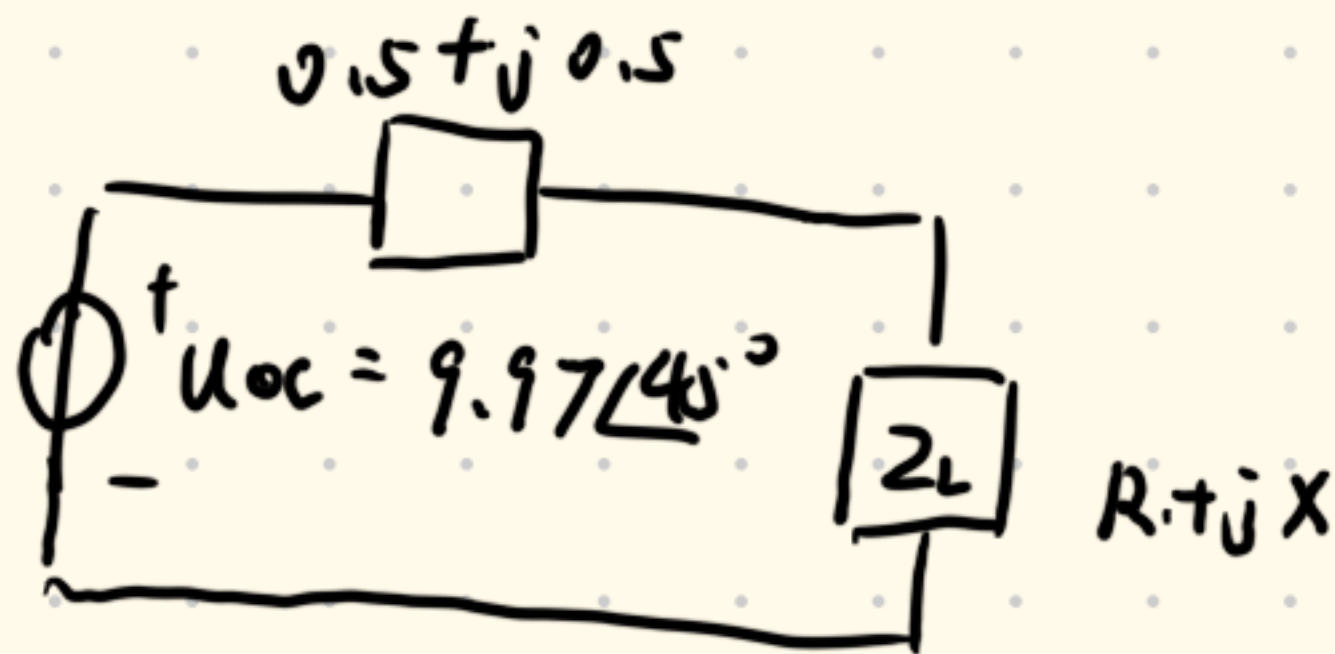
图题 9-14



图题 9-15

$$\dot{U}_{oc} = \frac{\dot{U}_1}{1+j1} \cdot 14.1 \angle 0^\circ = 10 \angle 45^\circ$$

$$Z = \frac{\dot{U}_1}{1+j1} = 0.5 + j0.5 \Omega$$



$$Z_L = 0.5 - j0.5 \Omega$$

Z_L 获得最大功率

$$P_{Lmax} = \frac{10^2}{4 \times 0.5} = 50 \text{ W}$$